Manikandan M

B.Tech Artificial Intelligence and Data Science
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Professional Summary

AI & Data Science undergraduate with a CGPA of 8.9 and three AI/ML internships spanning Agentic AI, Generative AI, and Machine Learning. Experienced in developing computer vision, machine learning, NLP, LLM, reinforcement learning, and multi-agent AI applications using Python. Hands-on experience with data preprocessing, model development, FastAPI, Streamlit, Gradio, REST APIs, and end-to-end AI/ML solutions.

Education

Er. Perumal Manimekalai College of Engineering, Hosur Oct 2023 - 2027 (Expected)
B.Tech, Artificial Intelligence and Data Science (Anna University) *CGPA: 8.9*
School: HSC (XII), 2021, 78.8% (Sri V.R.V Mat. Hr. Sec. School) | SSLC (X), 2019, 70.6% (Govt. Hr. Sec. School)

Technical Skills

- **Programming:** Python, Java, C, JavaScript, R
- **Machine Learning & AI:** Machine Learning, Supervised Learning, Regression, Classification, Data Preprocessing, Feature Engineering, EDA, Model Evaluation, Deep Learning, LSTM, Reinforcement Learning, Deep Q-Network (DQN), Computer Vision, Object Detection
- **Generative AI:** Generative AI, Large Language Models (LLMs), NLP, Prompt Engineering, Agentic AI, Multi-Agent AI
- **Libraries & Frameworks:** TensorFlow, PyTorch, Scikit-learn, XGBoost, YOLO11, OpenCV, NumPy, Pandas, NetworkX, Streamlit, Gradio, FastAPI, React
- **Web & APIs:** HTML, CSS, JavaScript, React, FastAPI, REST APIs, Uvicorn
- **Databases & Tools:** MySQL, SQLite, Git, GitHub, VS Code, Eclipse, Windows, Linux

Experience

Agentic AI Intern June 2026 - July 2026
Sisesoft IT Solutions *Hosur*

- Developed an autonomous AI Data Analyst agent using Python for automated CSV analysis and statistical processing.
- Integrated Python-based analysis tools with automated dashboard and report generation workflows.

Software Engineer Intern (AI & ML) December 2025 - January 2026
White And Box - Tech Products & Services *Bengaluru*

- Contributed to the Context Assembly Framework (CAF), an internal engineering tool for collecting and structuring contextual data.
- Developed Python backend logic and a Streamlit interface; managed JSON and SQLite storage and prepared data for machine learning workflows.
- Developed prediction and classification models and performed EDA, data preprocessing, and reproducible machine learning workflow implementation.

Generative AI Intern June 2025 - July 2025
Rubixe AI Solutions *Bengaluru*

- Applied Large Language Models (LLMs), NLP, and prompt engineering for automation and Generative AI use cases.
- Fine-tuned Generative AI models using NLP and deep learning techniques and developed prototype AI applications for internal solutions.

Projects

CapCell-AI - AI-Assisted Capsule Endoscopy Analysis
Python, YOLO11, PyTorch, OpenCV, FastAPI, Gradio, LLMs

- Developed a computer vision pipeline using YOLO11 for abnormality detection in capsule endoscopy images and converted detections into structured findings.
- Built a multi-agent LLM pipeline with report writer, recommendation, and review agents using the Groq API to generate structured clinical reports and consistency reviews.
- Developed a Gradio interface, FastAPI REST backend, and automated A4 PDF report generation with annotated images, findings, recommendations, and review outputs.

Quote Genie - Predictive Pricing Intelligence System
Python, XGBoost, FastAPI, React

- Developed XGBoost models for logistics price prediction and win-probability estimation using data preprocessing, feature engineering, and normalization.
- Developed an end-to-end prediction application using FastAPI backend and React frontend for model-driven pricing workflows.

AI Self-Healing Smart Grid
Python, NetworkX, LSTM, DQN, Reinforcement Learning

- Developed a smart-grid simulation using NetworkX with LSTM-based demand forecasting and DQN-based autonomous energy rerouting for fault recovery.
- Implemented a reinforcement learning agent for autonomous decision-making in simulated grid fault-recovery scenarios.

Certifications

- **Machine Learning and Deep Learning** - NPTEL (IIT Guwahati), Top 5%, Score: 72%
- **GenAI Powered Data Analytics Job Simulation** - Forage
- **Gen AI Exchange Hackathon** - Google Cloud
- **Microsoft Applied Skills: Generate reports with AI research agents**

Achievements & Leadership

- **1st Prize**, Code Debugging - PSV College of Engineering (2026)
- **2nd Prize**, Reverse Coding - Christ College of Science & Technology (2026)
- Student Coordinator - National-Level Conference and Department Symposium; Activity Coordinator - Department Tech Club